

Tue Jan 13

Nonlinear Dynamical Systems

$$\begin{aligned} \text{Ex: } F(x) &= 0 \quad x \in \mathbb{R}^n \\ F(x) &= Ax - b \quad : F \text{ is } \underline{\text{affine}} \text{ in } x \end{aligned}$$

EE510 : conditions on A and b :

$$\text{for existence and uniqueness of solution to } Ax=b \quad \begin{pmatrix} A \in \mathbb{R}^{n \times n} \\ b \in \mathbb{R}^{n \times 1} \end{pmatrix}$$

$$F(x) = Ax \quad : \text{linear}$$

$$F(x) = Ax + b \quad : \text{affine}$$

$$F(x) = x^T A x \quad : (\text{e.g. nonlinear})$$

Unconstrained optimization

$$\min_x g(x) \quad g: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\text{Necessary: } \begin{array}{l} \nabla g(x) = 0 \\ F(x) = 0 \end{array} \quad \left| \quad \begin{array}{l} g = \frac{1}{2} x^T A x - b x \\ \downarrow \\ \nabla g = Ax - b \end{array} \right.$$

F nonlinear: in general difficult to solve $F(x)=0$ analytically

Alternative: iterative approach

Gradient Descent

$$x^{k+1} = x^k - \underbrace{\alpha^k}_{\text{step-size}} \nabla g(x^k)$$

Discrete-time (DT)

$$\frac{x^{k+1} - x^k}{\alpha^k} = -\nabla g(x^k)$$

$$\frac{dx}{dt} = -\nabla g(x)$$

Continuous-time (CT)

$$\frac{dx}{dt} = -F(x)$$

$$\dot{x} = -F(x)$$



$$x^{k+1} = -\tilde{F}(x^k) \quad \times$$

$$\begin{aligned} \Rightarrow \frac{x^{k+1} - x^k}{\alpha} &= -F(x^k) \\ x^{k+1} &= \underbrace{x^k - \alpha F(x^k)}_{-\tilde{F}(x^k)} \end{aligned}$$

$$\therefore \tilde{F}(x^k) = \alpha F(x^k) - x^k$$

$$\frac{dx}{dt} = f(x) \quad x \in \mathbb{R}^n$$

t : time

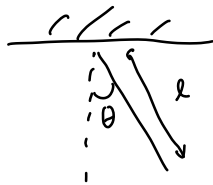
$$F: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \frac{dx}{dt} = \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \vdots \\ \dot{x}_n \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \vdots \\ \dot{x}_n \end{bmatrix} = f \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} f_1(x_1, \dots, x_n) \\ \vdots \\ f_n(x_1, \dots, x_n) \end{bmatrix}$$

$$\neq \begin{bmatrix} f_1(x_1) \\ \vdots \\ f_n(x_n) \end{bmatrix}$$

Ex.



m : mass

u : external force

① $u=0$

$$m l \frac{d^2 \theta}{dt^2} + l k \frac{d\theta}{dt} + m g \sin \theta = 0$$

$$\begin{cases} \theta = x_1 & \rightarrow \dot{\theta} = \dot{x}_1 = x_2 \\ \dot{\theta} = x_2 & \ddot{\theta} = \dot{x}_2 \end{cases}$$

$$m l \dot{x}_2 + l k x_2 + m g \sin x_1 = 0$$

$$\dot{x}_2 = -\frac{k}{m} x_2 - \frac{g}{l} \sin x_1$$

$$\dot{x}_1 = x_2$$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rightarrow X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$x_3 = t$$

$$\dot{x}_3 = 1$$

$$f_1(x_1, x_2) = x_2$$

$$f_2(x_1, x_2) = -\frac{k}{m} x_2 - \frac{g}{l} \sin x_1$$

$$f_1(x_1, x_2, x_3) = x_2$$

$$f_2(x_1, x_2, x_3) = -\frac{k}{m} x_2 - \frac{g}{l} \sin x_1$$

$$f_3(x_1, x_2, x_3) = 1$$

② $u = mg \cos \theta$

$$m l \frac{d^2 \theta}{dt^2} + l k \frac{d\theta}{dt} + mg \sin \theta = mg \cos \theta$$

$$m l \frac{d^2 \theta}{dt^2} + l k \frac{d\theta}{dt} + mg (\sin \theta - \cos \theta) = 0$$

$\Downarrow$

$$\dot{x} = f(x)$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \begin{array}{l} x_1 = \theta \\ x_2 = \dot{\theta} \end{array}$$

$$f_1(x_1, x_2) = x_2$$

$$f_2(x_1, x_2) = -\frac{k}{m} x_2 - \frac{g}{l} \sin x_1 + \frac{g}{l} \cos x_1$$

$$\dot{x} = f(x) : \text{no explicit } u$$

$\Rightarrow$ autonomous system

Forced system can still be autonomous!

③ $u = mg \cos(\omega t)$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{k}{m} x_2 - \frac{g}{l} \sin x_1 + \frac{g}{l} \cos(\omega t)$$

$$\begin{array}{l} \dot{x} = f(x) \quad - \text{time-invariant} \\ \dot{x} = f(x, t) \quad - \text{time-varying} \end{array} \quad \leftarrow \quad \text{define } \begin{bmatrix} x_3 = t \\ \dot{x}_3 = 1 \end{bmatrix}$$

$$\dot{x} = f(x, u)$$

$$m l \frac{d^2 \theta}{dt^2} + l k \frac{d\theta}{dt} + mg \sin \theta = u$$

$$\theta = x_1 \quad \dot{x}_1 = x_2 \quad = f_1(x_1, x_2, u)$$

$$\dot{\theta} = x_2 \quad \dot{x}_2 = -\frac{k}{m} x_2 - \frac{g}{l} \sin x_1 + u = f_2(x_1, x_2, u)$$

Linear
System

Specifying initial conditions

$$\dot{x} = f(x)$$

$$x(0) = x_0$$

$$\dot{x} = f(x, u)$$

$$x(0) = x_0$$

$$f: \underset{x}{\mathbb{R}^n} \times \underset{u}{\mathbb{R}^m} \rightarrow \mathbb{R}^n$$

$$\dot{x} = Ax$$

$$x(0) = x_0$$

$$f(x) = Ax$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$x \in \mathbb{R}^n \quad A \in \mathbb{R}^{n \times n}$$

$$\dot{x} = Ax + Bu$$

$$x(0) = x_0$$

$$f(x, u) = Ax + Bu$$

$$f: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$$

$$u \in \mathbb{R}^m \quad B \in \mathbb{R}^{n \times m}$$

$$x(t) = \underbrace{e^{At} x(0)}_{\text{I.C.}} + \underbrace{\int_0^t e^{A(t-\tau)} Bu(\tau) d\tau}_{\text{Input}}$$

$f(x)$ linear if:

$$- f(0) = 0$$

$$f(\alpha x) = \alpha f(x)$$

$$f(x_1 + x_2) = f(x_1) + f(x_2)$$

$$\dot{x} = -\alpha \frac{x^2}{k} = f(x)$$

$$\frac{\dot{z}}{2x} = -\alpha \frac{z}{k}$$

$$x^2 = z$$

$$\downarrow \text{differentiate}$$

$$2x \dot{x} = \dot{z}$$

$$\dot{x} = \frac{\dot{z}}{2x}$$

$$f(z) = -\alpha \frac{z}{k}$$

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix}$$

$$\dot{x} = -\alpha \frac{z}{k}$$

$$\dot{z} = -2\alpha \frac{xz}{k}$$

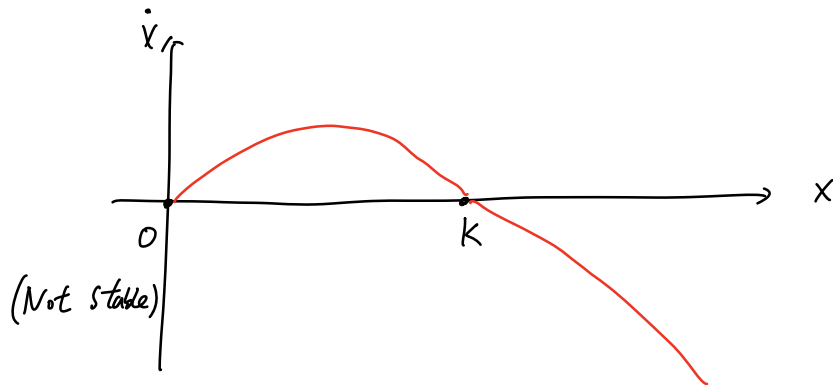
$$\dot{x} = \alpha \left(1 - \frac{x}{K}\right) x = f(x) \quad \text{with } \alpha > 0 \\ K > 0$$

$$f(x) = 0$$

$$\alpha \left(1 - \frac{x}{K}\right) x = 0$$

$$x = 0 \quad \text{and} \quad x = K$$

$x(0) = 0$ or K equilibrium points

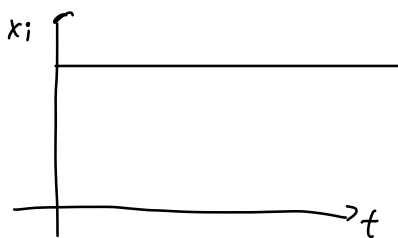


Equilibrium points :

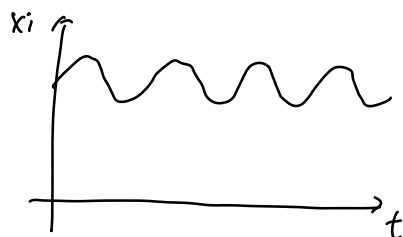
$$\dot{x} = f(x) \quad - (*)$$

Eq. pts. are constant trajectories

i.e. solution to (*) that does not change with time



vs.



$$f(x^*)=0 \Rightarrow x^* \text{ is an eq. pt}$$

$$f(x)=Ax$$

• Always has eq $x^*=0$

$$Ax=0$$

• Could have infinitely many eq, A is not invertible

$$A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

$$\dot{x} = -x$$

$x=0$ only eq. pt

$$\dot{x} = -x+t$$

$$f(x,t) \Rightarrow x=t$$

$$\dot{x} = -tx$$

$x=0$ only eq. pt

Suppose $\dot{x} = f(x)$ has unique eq of $x^*=k$

$$= x-k$$

$$\tilde{x} = x-k$$

$$\dot{\tilde{x}} = \dot{x}$$

$$= f(x)$$

$$= f(\tilde{x}+k)$$